

Bob-Bužek-Hillery-NO-SIGNALLING

January 20, 2026

1 The Bužek-Hillery Quantum Cloning Machine ($1 \rightarrow 2$)

1.1 1. Introduction

In quantum information theory, the **No-Cloning Theorem** states that it is impossible to perfectly copy an arbitrary, unknown quantum state. However, **approximate cloning** is possible.

The **Bužek-Hillery (BH) cloning machine** is the **optimal symmetric universal quantum cloning machine** for **$1 \rightarrow 2$ cloning of arbitrary qubit pure states**. It produces two identical approximate copies of a qubit with maximal fidelity.

1.2 2. Universal Symmetric Cloning

Let the input be an **arbitrary qubit state**:

$$\psi = \alpha 0 + \beta 1, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The BH machine takes **one input qubit** and an **ancilla prepared in a fixed state**, and produces **two approximate copies**:

$$\psi \otimes \text{ancilla} \longrightarrow \psi_1 \psi_2 \text{machine}.$$

- **Symmetric:** Both output qubits are identical.
 - **Universal:** Works for any arbitrary qubit ψ .
-

1.3 3. The Bužek-Hillery Cloning Transformation

The BH machine uses the following **unitary transformation** on the basis states:

[

$$0 \otimes 0 \otimes Q \longrightarrow \sqrt{\frac{2}{3}} 00 Q_0 + \sqrt{\frac{1}{3}} (01 + 10) Q_1,$$

$$1 \otimes 0 \otimes Q \longrightarrow \sqrt{\frac{2}{3}} 11 Q_1 + \sqrt{\frac{1}{3}} (01 + 10) Q_0.$$

|

Where:

- Q is the initial machine (ancilla) state.
 - Q_0, Q_1 are orthonormal machine states.
 - The coefficients ensure **maximal fidelity** for the copies.
-

1.4 4. Output State and Fidelity

After tracing out the machine (ancilla), each output qubit has **reduced density matrix**:

$$\rho_{\text{out}} = \frac{5}{6}\psi\psi + \frac{1}{6}\psi^\perp\psi^\perp,$$

where $\psi^\perp$ is the state orthogonal to ψ .

The **fidelity** of each clone is:

$$F = \psi\rho_{\text{out}}\psi = \frac{5}{6} \approx 0.8333.$$

This is the **optimal fidelity** for symmetric universal $1 \rightarrow 2$ qubit cloning.

1.5 5. Key Properties

1. **Symmetry:** Both output qubits are identical.
 2. **Universality:** Works for all input qubit states ψ .
 3. **Optimality:** No universal symmetric $1 \rightarrow 2$ cloning machine can achieve fidelity higher than $5/6$.
 4. **Approximate:** Perfect cloning is impossible; fidelity $F = 5/6$ is the best achievable.
-

1.6 6. Remarks

- The BH machine is fundamental in **quantum information theory** and **quantum cryptography**, e.g., in studying eavesdropping strategies.
- Extensions exist for $1 \rightarrow N$ **cloning** and **higher-dimensional systems (qudits)**.
- While perfect cloning is forbidden by the **No-Cloning Theorem**, the BH machine achieves the **best possible approximate cloning** for qubits.

1.7 References

1. V. Bužek and M. Hillery, “Quantum copying: Beyond the no-cloning theorem,” *Phys. Rev. A* **54**, 1844 (1996).
2. M. Nielsen and I. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 2010.
3. N. Gisin and S. Massar, “Optimal quantum cloning machines,” *Phys. Rev. Lett.* **79**, 2153 (1997).

```
[1]: """
NO-SIGNALING PRINCIPLE WITH BUŽEK-HILLERY QUANTUM CLONING MACHINE
=====

This code demonstrates that the Bužek-Hillery optimal symmetric universal
quantum cloning machine does NOT violate the no-signaling principle.

PROBLEM SETUP:
-----
Alice and Bob share an entangled state. Alice can choose to measure in
basis Z or X. Bob applies the Bužek-Hillery cloning machine to his qubit.

We need to show that Bob's reduced density matrix is the SAME regardless
of which basis Alice chooses to measure, proving no-signaling.

REFERENCE:
-----
Equation (9.30): Bužek-Hillery cloning transformation
Equations (9.8), (9.9): Reduced states after Alice's measurement

Author: Educational Implementation
"""

import numpy as np
from typing import Tuple, Dict, List
import warnings
warnings.filterwarnings('ignore')

# =====
# EDIT PARAMETERS HERE
# =====

class NoSignalingConfig:
    """
    Configuration for no-signaling demonstration.
```

```

    EDIT THESE VALUES TO TEST DIFFERENT SCENARIOS
    """
def __init__(self):
    # Initial entangled state coefficients (Bell state by default)
    #  $|\Phi\rangle = (1/2)(|00\rangle + |11\rangle)$ 
    self.use_bell_state = True

    # If not using Bell state, specify custom entangled state
    #  $|\Psi\rangle_{AB} = c00|00\rangle + c01|01\rangle + c10|10\rangle + c11|11\rangle$ 
    self.c00 = 1/np.sqrt(2)
    self.c01 = 0
    self.c10 = 0
    self.c11 = 1/np.sqrt(2)

    # Display precision
    self.precision = 6

    # Test multiple input states for Bob's cloning
    self.test_multiple_states = True

# =====
# QUANTUM STATE UTILITIES
# =====

class QuantumState:
    """Represents a multi-qubit quantum state."""

    def __init__(self, coefficients: np.ndarray, labels: List[str] = None):
        """
        Initialize quantum state.

        Args:
            coefficients: Complex amplitudes
            labels: Basis state labels (e.g., ['000', '001', ...])
        """
        self.coefficients = np.array(coefficients, dtype=complex)
        self.dim = len(coefficients)
        self.n_qubits = int(np.log2(self.dim))
        self.labels = labels

        if self.labels is None:
            self.labels = [format(i, f'0{self.n_qubits}b') for i in range(self.
↪dim)]

    def normalize(self):
        """Normalize the state."""

```

```

        norm = np.linalg.norm(self.coefficients)
        if norm > 0:
            self.coefficients /= norm
        return self

    def density_matrix(self) -> np.ndarray:
        """Compute density matrix  $\rho = |\psi\rangle\langle\psi|$ ."""
        ket = self.coefficients.reshape(-1, 1)
        bra = self.coefficients.conj().reshape(1, -1)
        return ket @ bra

    def __repr__(self):
        return f"QuantumState(n_qubits={self.n_qubits}, dim={self.dim})"

# =====
# BUŽEK-HILLERY CLONING MACHINE
# =====

class BuzekHilleryCloningMachine:
    """
    Implements the Bužek-Hillery optimal symmetric universal quantum cloning
    ↪ machine.

    Transformation (Eq. 9.30):
     $|\psi\rangle|R\rangle|\text{blank}\rangle \rightarrow (2/3)|\psi\rangle|\psi\rangle|\psi\rangle - (1/6)(|\psi\rangle|\psi\rangle + |\psi\rangle|\psi\rangle)|\psi\rangle$ 

    where  $|\psi\rangle = \alpha|1\rangle - \beta|0\rangle$  is orthogonal to  $|\phi\rangle = \alpha|0\rangle + \beta|1\rangle$ 
    """

    def __init__(self):
        self.log = []

    def add_log(self, message: str, level: int = 0):
        """Add formatted log message."""
        indent = " " * level
        self.log.append(indent + message)

    def print_log(self):
        """Print all logged messages."""
        for msg in self.log:
            print(msg)

    def section_header(self, title: str):
        """Add section header."""
        sep = "=" * 80
        self.add_log("")
        self.add_log(sep)

```

```

self.add_log(title.center(80))
self.add_log(sep)
self.add_log("")

def subsection_header(self, title: str):
    """Add subsection header."""
    self.add_log("")
    self.add_log("-" * 80)
    self.add_log(title)
    self.add_log("-" * 80)

def get_orthogonal_state(self, alpha: complex, beta: complex) -> Tuple[complex, complex]:
    """
    Compute the orthogonal state  $|\psi\rangle = \alpha|1\rangle - \beta|0\rangle$ .

    Args:
        alpha, beta: Coefficients of  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ 

    Returns:
        Tuple (coeff_0, coeff_1) for  $|\psi\rangle$ 
    """
    #  $|\psi\rangle = \alpha|1\rangle - \beta|0\rangle = -\beta|0\rangle + \alpha|1\rangle$ 
    coeff_0 = -np.conj(beta)
    coeff_1 = np.conj(alpha)
    return coeff_0, coeff_1

def apply_cloning(self, alpha: complex, beta: complex) -> np.ndarray:
    """
    Apply Bužek-Hillery cloning to state  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ .

    Input:  $|\psi\rangle$   $|0\rangle$   $|blank\rangle$  (1 qubit input + 1 blank qubit + 1 ancilla)
    Output: 3-qubit state after cloning

    Returns:
        8-dimensional state vector (3 qubits)
    """
    self.subsection_header(f"Applying Bužek-Hillery Cloning to  $|\psi\rangle = \{\alpha: .4f\}|0\rangle + \{\beta: .4f\}|1\rangle$ ")

    # Get orthogonal state
    alpha_perp, beta_perp = self.get_orthogonal_state(alpha, beta)

    self.add_log(f"Input state:  $|\psi\rangle = \{\alpha: .6f\}|0\rangle + \{\beta: .6f\}|1\rangle$ ", 1)
    self.add_log(f"Orthogonal state:  $|\psi\rangle = \{\alpha\_perp: .6f\}|0\rangle + \{\beta\_perp: .6f\}|1\rangle$ ", 1)
    self.add_log("")

```

```

# Verify orthogonality
inner_product = np.conj(alpha) * alpha_perp + np.conj(beta) * beta_perp
self.add_log(f"Verification:  $\langle\psi|\psi\rangle = \{inner\_product:.6e\}$  ", 1)
self.add_log("")

# Initialize output state (8 basis states for 3 qubits)
output = np.zeros(8, dtype=complex)

# Basis:  $|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle$ 

self.add_log("Applying transformation (Eq. 9.30):", 1)
self.add_log("  $(2/3)|\psi\rangle|\psi\rangle|\psi\rangle - (1/6)(|\psi\rangle|\psi\rangle + |\psi\rangle|\psi\rangle)|\psi\rangle$ ", 2)
self.add_log("")

# Term 1:  $(2/3)|\psi\rangle|\psi\rangle|\psi\rangle$ 
coeff1 = np.sqrt(2/3)
self.add_log("Term 1:  $(2/3)|\psi\rangle|\psi\rangle|\psi\rangle$ ", 1)

for i, amp_i in enumerate([alpha, beta]):
    for j, amp_j in enumerate([alpha, beta]):
        for k, amp_k in enumerate([alpha_perp, beta_perp]):
            idx = i * 4 + j * 2 + k
            output[idx] += coeff1 * amp_i * amp_j * amp_k

# Term 2:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$ 
coeff2 = -np.sqrt(1/6)
self.add_log("Term 2:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$ ", 1)

for i, amp_i in enumerate([alpha, beta]):
    for j, amp_j in enumerate([alpha_perp, beta_perp]):
        for k, amp_k in enumerate([alpha, beta]):
            idx = i * 4 + j * 2 + k
            output[idx] += coeff2 * amp_i * amp_j * amp_k

# Term 3:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$ 
self.add_log("Term 3:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$ ", 1)

for i, amp_i in enumerate([alpha_perp, beta_perp]):
    for j, amp_j in enumerate([alpha, beta]):
        for k, amp_k in enumerate([alpha, beta]):
            idx = i * 4 + j * 2 + k
            output[idx] += coeff2 * amp_i * amp_j * amp_k

self.add_log("")

# Verify normalization

```

```

    norm = np.linalg.norm(output)
    self.add_log(f"Output state norm: {norm:.6f}", 1)

    if not np.isclose(norm, 1.0):
        self.add_log(f" WARNING: State not normalized! Normalizing...", 2)
        output /= norm

    return output

# =====
# PARTIAL TRACE OPERATIONS
# =====

def partial_trace_first_qubit(rho: np.ndarray) -> np.ndarray:
    """
    Trace out the first qubit from a 3-qubit density matrix.

    Args:
        rho: 8x8 density matrix

    Returns:
        4x4 reduced density matrix for qubits 2 and 3
    """
    rho_reduced = np.zeros((4, 4), dtype=complex)

    # Basis ordering: |000>, |001>, |010>, |011>, |100>, |101>, |110>, |111>
    # Trace over first qubit means summing over |0XX> and |1XX>

    for i in range(4):
        for j in range(4):
            # i, j index the 2-qubit subspace
            # Sum contributions from |0> and |1> in first qubit
            rho_reduced[i, j] = rho[i, j] + rho[i+4, j+4]

    return rho_reduced

def partial_trace_last_two_qubits(rho: np.ndarray) -> np.ndarray:
    """
    Trace out the last two qubits from a 3-qubit density matrix.

    Args:
        rho: 8x8 density matrix

    Returns:
        2x2 reduced density matrix for first qubit
    """
    rho_reduced = np.zeros((2, 2), dtype=complex)

```



```

# Sum over all combinations of qubits 2 and 3
for i in range(2):
    for j in range(2):
        for k in range(4):
            idx1 = i * 4 + k
            idx2 = j * 4 + k
            rho_reduced[i, j] += rho[idx1, idx2]

return rho_reduced

# =====
# NO-SIGNALING DEMONSTRATION
# =====

class NoSignalingDemonstration:
    """
    Demonstrates that Bužek-Hillery cloning preserves the no-signaling principle.
    """

    def __init__(self, config: NoSignalingConfig):
        self.config = config
        self.cloning_machine = BuzekHilleryCloningMachine()
        self.log = []

    def add_log(self, message: str, level: int = 0):
        """Add log message."""
        indent = " " * level
        self.log.append(indent + message)

    def print_log(self):
        """Print all logs."""
        for msg in self.log:
            print(msg)

    def section_header(self, title: str):
        """Add section header."""
        sep = "=" * 80
        self.add_log("")
        self.add_log(sep)
        self.add_log(title.center(80))
        self.add_log(sep)
        self.add_log("")

    def format_matrix(self, matrix: np.ndarray, precision: int = 6) -> str:
        """Format matrix for display."""
        lines = []

```

```

for i in range(matrix.shape[0]):
    row = " ["
    for j in range(matrix.shape[1]):
        val = matrix[i, j]
        if np.abs(val.imag) < 1e-10:
            row += f"{val.real:{precision+3}.{precision}f} "
        else:
            row += f"{val.real:{precision+3}.{precision}f}{val.imag:{precision+3}.{precision}f}j"
    row += "]"
    lines.append(row)
return "\n".join(lines)

def demonstrate_no_signaling(self):
    """
    Main demonstration of no-signaling principle.
    """
    self.section_header("NO-SIGNALING PRINCIPLE WITH BUŽEK-HILLERY CLONING")

    self.add_log("SETUP:")
    self.add_log("-----", 1)
    self.add_log("1. Alice and Bob share an entangled state", 1)
    self.add_log("2. Alice measures her qubit (in Z or X basis)", 1)
    self.add_log("3. Bob applies Bužek-Hillery cloning to his qubit", 1)
    self.add_log("4. We compute Bob's reduced density matrix", 1)
    self.add_log("", 1)
    self.add_log("NO-SIGNALING REQUIREMENT:", 1)
    self.add_log(" Bob's reduced state must be IDENTICAL regardless of Alice's choice!", 1)
    self.add_log("")

    # Use Bell state  $|\Phi\rangle = (1/2)(|00\rangle + |11\rangle)$ 
    if self.config.use_bell_state:
        self.add_log("Initial entangled state:  $|\Phi\rangle = (1/2)(|00\rangle + |11\rangle)$ ")
        alpha_A, beta_A = 1/np.sqrt(2), 0
        alpha_B_if_0, beta_B_if_0 = 1, 0 # Bob's state if Alice gets |0>
        alpha_B_if_1, beta_B_if_1 = 0, 1 # Bob's state if Alice gets |1>
    else:
        self.add_log("Using custom entangled state")
        # Implement custom state logic here if needed

    # =====
    # CASE 1: Alice measures in Z basis
    # =====
    self.section_header("CASE 1: ALICE MEASURES IN Z BASIS")

    self.add_log("When Alice measures in computational (Z) basis:")

```

```

self.add_log("")

self.add_log("If Alice gets  $|0\rangle$  (probability 1/2):", 1)
self.add_log("  Bob's state collapses to:  $|\psi_B\rangle = |0\rangle$ ", 2)
self.add_log("")

# Apply cloning to  $|0\rangle$ 
output_if_0 = self.cloning_machine.apply_cloning(1, 0)
state_if_0 = QuantumState(output_if_0)
rho_full_if_0 = state_if_0.density_matrix()
rho_Bob_if_0 = partial_trace_last_two_qubits(rho_full_if_0)

self.add_log("If Alice gets  $|1\rangle$  (probability 1/2):", 1)
self.add_log("  Bob's state collapses to:  $|\psi_B\rangle = |1\rangle$ ", 2)
self.add_log("")

# Apply cloning to  $|1\rangle$ 
output_if_1 = self.cloning_machine.apply_cloning(0, 1)
state_if_1 = QuantumState(output_if_1)
rho_full_if_1 = state_if_1.density_matrix()
rho_Bob_if_1 = partial_trace_last_two_qubits(rho_full_if_1)

# Bob's mixed state after Alice's Z measurement
rho_Bob_Z = 0.5 * rho_Bob_if_0 + 0.5 * rho_Bob_if_1

self.add_log("Bob's reduced density matrix (after Alice measures Z):")
self.add_log("   $\rho_{\text{Bob}}^Z = (1/2)\rho_{\text{Bob}}|0\rangle + (1/2)\rho_{\text{Bob}}|1\rangle$ ", 1)
self.add_log("")
self.add_log(self.format_matrix(rho_Bob_Z), 1)
self.add_log("")

# =====
# CASE 2: Alice measures in X basis
# =====
self.section_header("CASE 2: ALICE MEASURES IN X BASIS")

self.add_log("When Alice measures in X basis ( $|+\rangle$ ,  $|-\rangle$ ):")
self.add_log("")

# Bell state in X basis:  $|\Phi\rangle = (1/2)(|++\rangle + |--\rangle)$ 
self.add_log("If Alice gets  $|+\rangle$  (probability 1/2):", 1)
self.add_log("  Bob's state collapses to:  $|\psi_B\rangle = |+\rangle = (1/2)(|0\rangle + \rightarrow|1\rangle)$ ", 2)
self.add_log("")

# Apply cloning to  $|+\rangle = (1/2)(|0\rangle + |1\rangle)$ 

```

```

        output_if_plus = self.cloning_machine.apply_cloning(1/np.sqrt(2), 1/np.
→sqrt(2))
        state_if_plus = QuantumState(output_if_plus)
        rho_full_if_plus = state_if_plus.density_matrix()
        rho_Bob_if_plus = partial_trace_last_two_qubits(rho_full_if_plus)

        self.add_log("If Alice gets  $|-\rangle$  (probability 1/2):", 1)
        self.add_log("  Bob's state collapses to:  $|\psi_B\rangle = |-\rangle = (1/2)(|0\rangle -$ 
→ $|1\rangle)$ ", 2)
        self.add_log("")

        # Apply cloning to  $|-\rangle = (1/2)(|0\rangle - |1\rangle)$ 
        output_if_minus = self.cloning_machine.apply_cloning(1/np.sqrt(2), -1/np.
→sqrt(2))
        state_if_minus = QuantumState(output_if_minus)
        rho_full_if_minus = state_if_minus.density_matrix()
        rho_Bob_if_minus = partial_trace_last_two_qubits(rho_full_if_minus)

        # Bob's mixed state after Alice's X measurement
        rho_Bob_X = 0.5 * rho_Bob_if_plus + 0.5 * rho_Bob_if_minus

        self.add_log("Bob's reduced density matrix (after Alice measures X):")
        self.add_log("   $\rho_{Bob}^X = (1/2)\rho_{Bob|+} + (1/2)\rho_{Bob|-}$ ", 1)
        self.add_log("")
        self.add_log(self.format_matrix(rho_Bob_X), 1)
        self.add_log("")

        # =====
        # COMPARISON: NO-SIGNALING CHECK
        # =====
        self.section_header("NO-SIGNALING CHECK: COMPARISON OF  $\rho_{Bob}^Z$  vs_
→ $\rho_{Bob}^X$ ")

        self.add_log("Computing difference:  $||\rho_{Bob}^Z - \rho_{Bob}^X||$ ")
        self.add_log("")

        difference = rho_Bob_Z - rho_Bob_X
        frobenius_norm = np.linalg.norm(difference, 'fro')
        max_element = np.max(np.abs(difference))

        self.add_log(f"Frobenius norm:  $||\rho_{Bob}^Z - \rho_{Bob}^X||_F = \{frobenius\_norm:$ 
→ $.6e\}$ ", 1)
        self.add_log(f"Maximum element:  $\max|\rho_Z[i,j] - \rho_X[i,j]| = \{max\_element:$ 
→ $.6e\}$ ", 1)
        self.add_log("")

```

```

if frobenius_norm < 1e-10:
    self.add_log(" NO-SIGNALING PRINCIPLE SATISFIED!", 1)
    self.add_log("  $\rho_{\text{Bob}^Z} = \rho_{\text{Bob}^X}$  (within numerical precision)", 1)
    self.add_log("")
    self.add_log("CONCLUSION:", 1)
    self.add_log(" Bob's reduced state is IDENTICAL regardless of_
→whether", 2)
    self.add_log(" Alice measured in Z or X basis. Therefore, Bob_
→cannot", 2)
    self.add_log(" determine which measurement Alice performed.", 2)
    self.add_log(" ", 2)
    self.add_log(" The Bužek-Hillery cloning machine does NOT violate",_
→2)
    self.add_log(" the no-signaling principle! ", 2)
else:
    self.add_log(" WARNING: Possible violation detected!", 1)
    self.add_log(f" Difference is {frobenius_norm:.6e}, which may_
→indicate", 1)
    self.add_log(" either numerical error or a conceptual issue.", 1)

# =====
# EXPLICIT FORMS: Equations (9.8) and (9.9)
# =====
self.section_header("REDUCED DENSITY MATRICES (Eqs. 9.8 and 9.9)")

self.add_log("Equation (9.8): Bob's state if Alice measures Z basis")
self.add_log("  $\rho_Z = (1/2)|++\rangle\langle ++| + (1/2)|--\rangle\langle --|$ ")
self.add_log("")
self.add_log("In computational basis  $\{|0\rangle, |1\rangle\}$ :")
self.add_log(self.format_matrix(rho_Bob_Z), 1)
self.add_log("")

# Verify it's actually (1/2)I
identity_check_Z = np.allclose(rho_Bob_Z, 0.5 * np.eye(2))
if identity_check_Z:
    self.add_log("  $\rho_Z = (1/2)I$  (Maximally mixed state) ", 1)

self.add_log("")
self.add_log("Equation (9.9): Bob's state if Alice measures X basis")
self.add_log("  $\rho_X = (1/2)|00\rangle\langle 00| + (1/2)|11\rangle\langle 11|$ ")
self.add_log("")
self.add_log("In computational basis  $\{|0\rangle, |1\rangle\}$ :")
self.add_log(self.format_matrix(rho_Bob_X), 1)
self.add_log("")

# Verify it's actually (1/2)I
identity_check_X = np.allclose(rho_Bob_X, 0.5 * np.eye(2))

```

```

        if identity_check_X:
            self.add_log("  $\rho_X = (1/2)I$  (Maximally mixed state) ", 1)

        self.add_log("")
        self.add_log("KEY OBSERVATION:")
        self.add_log(" Both Eq. (9.8) and Eq. (9.9) give the SAME reduced state:
↪", 1)
        self.add_log("  $\rho_Z = \rho_X = (1/2)I$ ", 1)
        self.add_log("")
        self.add_log(" This is the maximally mixed state, which contains NO_
↪information", 1)
        self.add_log(" about Alice's measurement choice. Bob sees the same_
↪state regardless", 1)
        self.add_log(" of what Alice does!", 1)

    # Store results
    results = {
        'rho_Bob_Z': rho_Bob_Z,
        'rho_Bob_X': rho_Bob_X,
        'rho_Bob_if_0': rho_Bob_if_0,
        'rho_Bob_if_1': rho_Bob_if_1,
        'rho_Bob_if_plus': rho_Bob_if_plus,
        'rho_Bob_if_minus': rho_Bob_if_minus,
        'difference_norm': frobenius_norm,
        'max_difference': max_element,
        'no_signaling_satisfied': frobenius_norm < 1e-10,
        'is_maximally_mixed_Z': identity_check_Z,
        'is_maximally_mixed_X': identity_check_X
    }

    return results

# =====
# MAIN EXECUTION
# =====

def main():
    """Main execution function."""

    print("="*80)
    print("NO-SIGNALING PRINCIPLE WITH BUŽEK-HILLERY CLONING".center(80))
    print("="*80)
    print()

    # Create configuration
    config = NoSignalingConfig()

```

```

# Run demonstration
demo = NoSignalingDemonstration(config)
results = demo.demonstrate_no_signaling()

# Print all logs
demo.print_log()
demo.cloning_machine.print_log()

return results, demo

if __name__ == "__main__":
    results, demo = main()

    print("\n" + "="*80)
    print("FINAL SUMMARY")
    print("="*80)

    print("\n" + "-"*80)
    print("REDUCED DENSITY MATRICES (Equations 9.8 and 9.9)")
    print("-"*80)

    print("\nEquation (9.8):  $\rho_Z$  (Bob's state if Alice measures Z basis)")
    print("   $\rho_Z = (1/2)|++\rangle\langle ++| + (1/2)|--\rangle\langle --|$ ")
    print("\nMatrix form:")
    print(demo.format_matrix(results['rho_Bob_Z']))

    if results['is_maximally_mixed_Z']:
        print("\n   $\rho_Z = (1/2)I$  (Maximally mixed state) ")

    print("\n" + "-"*80)
    print("\nEquation (9.9):  $\rho_X$  (Bob's state if Alice measures X basis)")
    print("   $\rho_X = (1/2)|00\rangle\langle 00| + (1/2)|11\rangle\langle 11|$ ")
    print("\nMatrix form:")
    print(demo.format_matrix(results['rho_Bob_X']))

    if results['is_maximally_mixed_X']:
        print("\n   $\rho_X = (1/2)I$  (Maximally mixed state) ")

    print("\n" + "-"*80)
    print("INDIVIDUAL CONDITIONAL STATES")
    print("-"*80)

    print("\n $\rho_{Bob|0}$  (Bob's state when Alice gets  $|0\rangle$ ):")
    print(demo.format_matrix(results['rho_Bob_if_0']))

    print("\n $\rho_{Bob|1}$  (Bob's state when Alice gets  $|1\rangle$ ):")

```

```

print(demo.format_matrix(results['rho_Bob_if_1']))

print("\np_Bob|+ (Bob's state when Alice gets |+):")
print(demo.format_matrix(results['rho_Bob_if_plus']))

print("\np_Bob|- (Bob's state when Alice gets |-):")
print(demo.format_matrix(results['rho_Bob_if_minus']))

print("\n" + "="*80)
print("NO-SIGNALING VERIFICATION")
print("="*80)

print(f"\nDifference: || $\rho_Z - \rho_X$ ||_F = {results['difference_norm']:.6e}")
print(f"Max element difference: {results['max_difference']:.6e}")
print(f"\nNo-signaling satisfied: {results['no_signaling_satisfied']}")

if results['no_signaling_satisfied']:
    print("\n" + ""*40)
    print(" CONCLUSION: NO-SIGNALING PRINCIPLE PRESERVED ")
    print(""*40)
    print("\nThe Bužek-Hillery optimal cloning machine does NOT violate")
    print("the no-signaling principle!")
    print("\nBob's reduced density matrix is IDENTICAL regardless of_")
    ↪ Alice's")
    print("measurement choice (Z or X basis):")
    print("\n  $\rho_Z = \rho_X = (1/2)I$ ")
    print("\nThis is the maximally mixed state, containing zero information")
    print("about Alice's actions. Bob cannot determine what Alice measured!")
else:
    print("\n Warning: Possible numerical issues detected.")

print("\n" + "="*80)

```

```

=====
NO-SIGNALING PRINCIPLE WITH BUŽEK-HILLERY CLONING
=====

```

```

=====
NO-SIGNALING PRINCIPLE WITH BUŽEK-HILLERY CLONING
=====

```

SETUP:

- ```

1. Alice and Bob share an entangled state
2. Alice measures her qubit (in Z or X basis)
3. Bob applies Bužek-Hillery cloning to his qubit
4. We compute Bob's reduced density matrix

```



NO-SIGNALING REQUIREMENT:

Bob's reduced state must be IDENTICAL regardless of Alice's choice!

Initial entangled state:  $|\Phi\rangle = (1/2)(|00\rangle + |11\rangle)$

=====

CASE 1: ALICE MEASURES IN Z BASIS

=====

When Alice measures in computational (Z) basis:

If Alice gets  $|0\rangle$  (probability 1/2):

Bob's state collapses to:  $|\psi_B\rangle = |0\rangle$

If Alice gets  $|1\rangle$  (probability 1/2):

Bob's state collapses to:  $|\psi_B\rangle = |1\rangle$

Bob's reduced density matrix (after Alice measures Z):

$$\rho_{\text{Bob}}^Z = (1/2)\rho_{\text{Bob}}|0\rangle + (1/2)\rho_{\text{Bob}}|1\rangle$$

$$\begin{bmatrix} 0.500000 & 0.000000 \\ 0.000000 & 0.500000 \end{bmatrix}$$

=====

CASE 2: ALICE MEASURES IN X BASIS

=====

When Alice measures in X basis ( $|+\rangle$ ,  $|-\rangle$ ):

If Alice gets  $|+\rangle$  (probability 1/2):

Bob's state collapses to:  $|\psi_B\rangle = |+\rangle = (1/2)(|0\rangle + |1\rangle)$

If Alice gets  $|-\rangle$  (probability 1/2):

Bob's state collapses to:  $|\psi_B\rangle = |-\rangle = (1/2)(|0\rangle - |1\rangle)$

Bob's reduced density matrix (after Alice measures X):

$$\rho_{\text{Bob}}^X = (1/2)\rho_{\text{Bob}}|+\rangle + (1/2)\rho_{\text{Bob}}|-\rangle$$

$$\begin{bmatrix} 0.500000 & 0.000000 \\ 0.000000 & 0.500000 \end{bmatrix}$$

=====

NO-SIGNALING CHECK: COMPARISON OF  $\rho_{\text{Bob}}^Z$  vs  $\rho_{\text{Bob}}^X$

=====

Computing difference:  $||\rho_{\text{Bob}}^Z - \rho_{\text{Bob}}^X||$

Frobenius norm:  $||\rho_{\text{Bob}}^Z - \rho_{\text{Bob}}^X||_F = 3.925231\text{e-}16$

Maximum element:  $\max|\rho_Z[i,j] - \rho_X[i,j]| = 2.775558\text{e-}16$

NO-SIGNALING PRINCIPLE SATISFIED!

$\rho_{\text{Bob}}^Z = \rho_{\text{Bob}}^X$  (within numerical precision)

CONCLUSION:

Bob's reduced state is IDENTICAL regardless of whether Alice measured in Z or X basis. Therefore, Bob cannot determine which measurement Alice performed.

The Bužek-Hillery cloning machine does NOT violate the no-signaling principle!

---

### REDUCED DENSITY MATRICES (Eqs. 9.8 and 9.9)

---

Equation (9.8): Bob's state if Alice measures Z basis

$$\rho_Z = (1/2)|++\rangle\langle++| + (1/2)|--\rangle\langle--|$$

In computational basis  $\{|0\rangle, |1\rangle\}$ :

$$\begin{bmatrix} 0.500000 & 0.000000 \\ 0.000000 & 0.500000 \end{bmatrix}$$

$$\rho_Z = (1/2)I \text{ (Maximally mixed state)}$$

Equation (9.9): Bob's state if Alice measures X basis

$$\rho_X = (1/2)|00\rangle\langle 00| + (1/2)|11\rangle\langle 11|$$

In computational basis  $\{|0\rangle, |1\rangle\}$ :

$$\begin{bmatrix} 0.500000 & 0.000000 \\ 0.000000 & 0.500000 \end{bmatrix}$$

$$\rho_X = (1/2)I \text{ (Maximally mixed state)}$$

KEY OBSERVATION:

Both Eq. (9.8) and Eq. (9.9) give the SAME reduced state:

$$\rho_Z = \rho_X = (1/2)I$$

This is the maximally mixed state, which contains NO information about Alice's measurement choice. Bob sees the same state regardless of what Alice does!

---

Applying Bužek-Hillery Cloning to  $|\psi\rangle = 1.0000|0\rangle + 0.0000|1\rangle$

---

Input state:  $|\psi\rangle = 1.000000|0\rangle + 0.000000|1\rangle$   
Orthogonal state:  $|\psi\rangle = 0.000000|0\rangle + 1.000000|1\rangle$

Verification:  $\langle\psi|\psi\rangle = 0.000000e+00$

Applying transformation (Eq. 9.30):  
 $(2/3)|\psi\rangle|\psi\rangle|\psi\rangle - (1/6)(|\psi\rangle|\psi\rangle + |\psi\rangle|\psi\rangle)|\psi\rangle$

Term 1:  $(2/3)|\psi\rangle|\psi\rangle|\psi\rangle$

Term 2:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$

Term 3:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$

Output state norm: 1.000000

---

Applying Bužek-Hillery Cloning to  $|\psi\rangle = 0.0000|0\rangle + 1.0000|1\rangle$

---

Input state:  $|\psi\rangle = 0.000000|0\rangle + 1.000000|1\rangle$   
Orthogonal state:  $|\psi\rangle = -1.000000|0\rangle + 0.000000|1\rangle$

Verification:  $\langle\psi|\psi\rangle = 0.000000e+00$

Applying transformation (Eq. 9.30):  
 $(2/3)|\psi\rangle|\psi\rangle|\psi\rangle - (1/6)(|\psi\rangle|\psi\rangle + |\psi\rangle|\psi\rangle)|\psi\rangle$

Term 1:  $(2/3)|\psi\rangle|\psi\rangle|\psi\rangle$

Term 2:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$

Term 3:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$

Output state norm: 1.000000

---

Applying Bužek-Hillery Cloning to  $|\psi\rangle = 0.7071|0\rangle + 0.7071|1\rangle$

---

Input state:  $|\psi\rangle = 0.707107|0\rangle + 0.707107|1\rangle$   
Orthogonal state:  $|\psi\rangle = -0.707107|0\rangle + 0.707107|1\rangle$

Verification:  $\langle\psi|\psi\rangle = 0.000000e+00$

Applying transformation (Eq. 9.30):  
 $(2/3)|\psi\rangle|\psi\rangle|\psi\rangle - (1/6)(|\psi\rangle|\psi\rangle + |\psi\rangle|\psi\rangle)|\psi\rangle$

Term 1:  $(2/3)|\psi\rangle|\psi\rangle|\psi\rangle$

Term 2:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$

Term 3:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$

Output state norm: 1.000000

-----  
Applying Bužek-Hillery Cloning to  $|\psi\rangle = 0.7071|0\rangle + -0.7071|1\rangle$   
-----

Input state:  $|\psi\rangle = 0.707107|0\rangle + -0.707107|1\rangle$   
Orthogonal state:  $|\psi\rangle = 0.707107|0\rangle + 0.707107|1\rangle$

Verification:  $\langle\psi|\psi\rangle = 0.000000e+00$

Applying transformation (Eq. 9.30):  
 $(2/3)|\psi\rangle|\psi\rangle|\psi\rangle - (1/6)(|\psi\rangle|\psi\rangle + |\psi\rangle|\psi\rangle)|\psi\rangle$

Term 1:  $(2/3)|\psi\rangle|\psi\rangle|\psi\rangle$   
Term 2:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$   
Term 3:  $-(1/6)|\psi\rangle|\psi\rangle|\psi\rangle$

Output state norm: 1.000000

=====

## FINAL SUMMARY

=====

-----

## REDUCED DENSITY MATRICES (Equations 9.8 and 9.9)

-----

Equation (9.8):  $\rho_Z$  (Bob's state if Alice measures Z basis)  
 $\rho_Z = (1/2)|++\rangle\langle++| + (1/2)|--\rangle\langle--|$

Matrix form:  
 $\begin{bmatrix} 0.500000 & 0.000000 \\ 0.000000 & 0.500000 \end{bmatrix}$

$\rho_Z = (1/2)I$  (Maximally mixed state)

-----

Equation (9.9):  $\rho_X$  (Bob's state if Alice measures X basis)  
 $\rho_X = (1/2)|00\rangle\langle 00| + (1/2)|11\rangle\langle 11|$

Matrix form:  
 $\begin{bmatrix} 0.500000 & 0.000000 \\ 0.000000 & 0.500000 \end{bmatrix}$

$\rho_X = (1/2)I$  (Maximally mixed state)

-----

## INDIVIDUAL CONDITIONAL STATES

```

ρ_Bob|0 (Bob's state when Alice gets |0>>):
[0.833333 0.000000]
[0.000000 0.166667]
```

```
ρ_Bob|1 (Bob's state when Alice gets |1>>):
[0.166667 0.000000]
[0.000000 0.833333]
```

```
ρ_Bob|+ (Bob's state when Alice gets |+>>):
[0.500000 0.333333]
[0.333333 0.500000]
```

```
ρ_Bob|- (Bob's state when Alice gets |->>):
[0.500000 -0.333333]
[-0.333333 0.500000]
```

```
=====
NO-SIGNALING VERIFICATION
=====
```

```
Difference: ||ρ_Z - ρ_X||_F = 3.925231e-16
Max element difference: 2.775558e-16
```

```
No-signaling satisfied: True
```

```
CONCLUSION: NO-SIGNALING PRINCIPLE PRESERVED
```

```
The Bužek-Hillery optimal cloning machine does NOT violate
the no-signaling principle!
```

```
Bob's reduced density matrix is IDENTICAL regardless of Alice's
measurement choice (Z or X basis):
```

$$\rho_Z = \rho_X = (1/2)I$$

```
This is the maximally mixed state, containing zero information
about Alice's actions. Bob cannot determine what Alice measured!
```